

Module 3

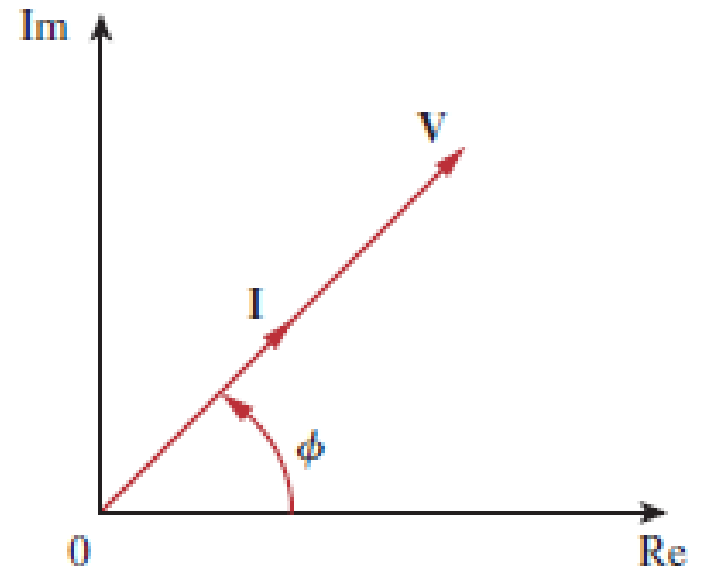
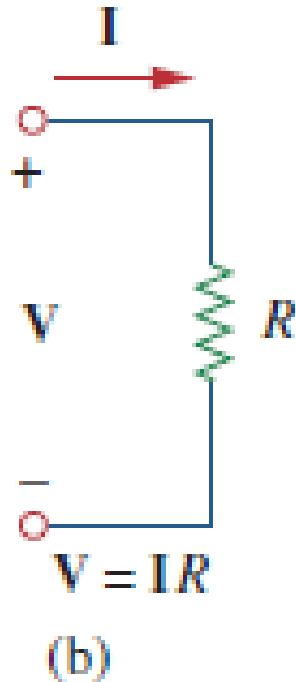
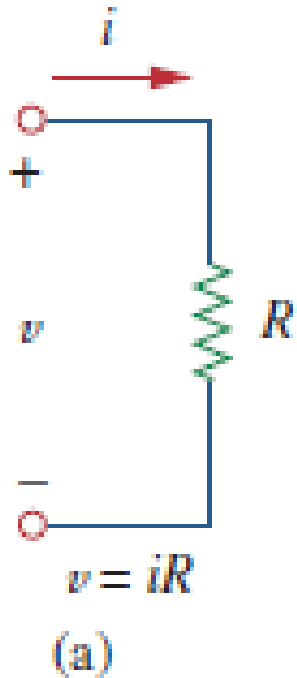
Sinusoidal Steady-State Analysis

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Phasor relationship for circuit elements

RESISTOR



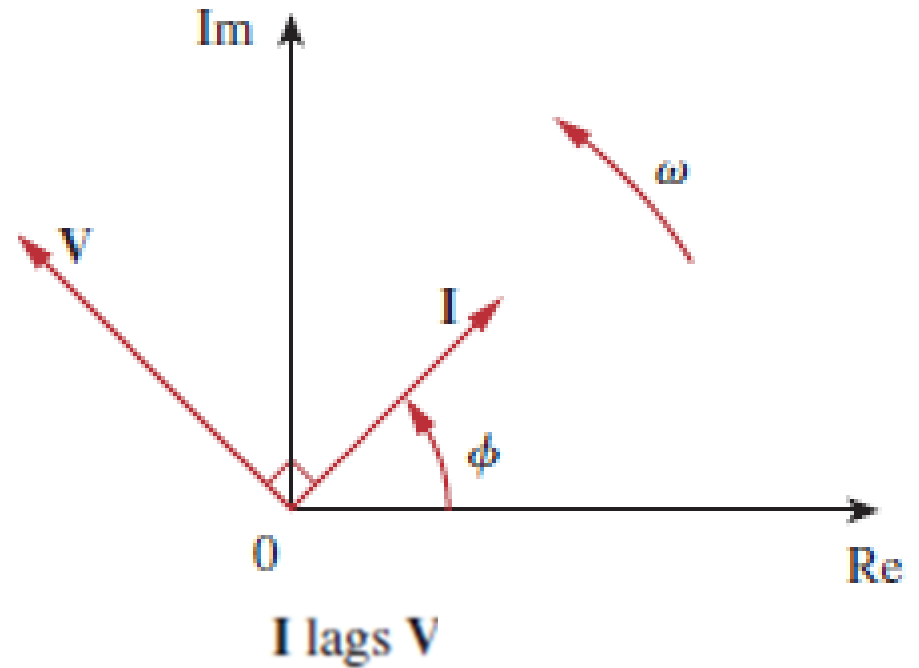
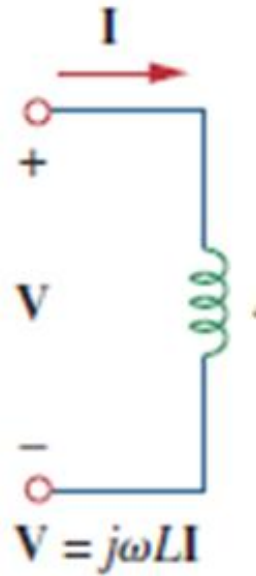
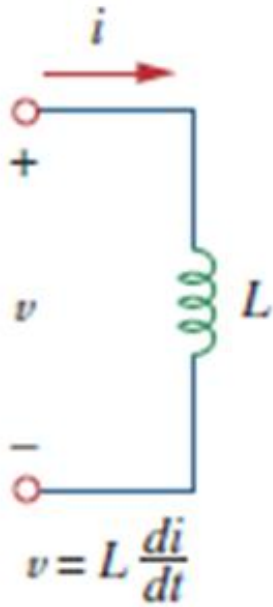
$$v = iR = RI_m \cos(\omega t + \phi)$$

$$\mathbf{V} = RI_m \angle \phi$$

$$\mathbf{V} = R\mathbf{I}$$

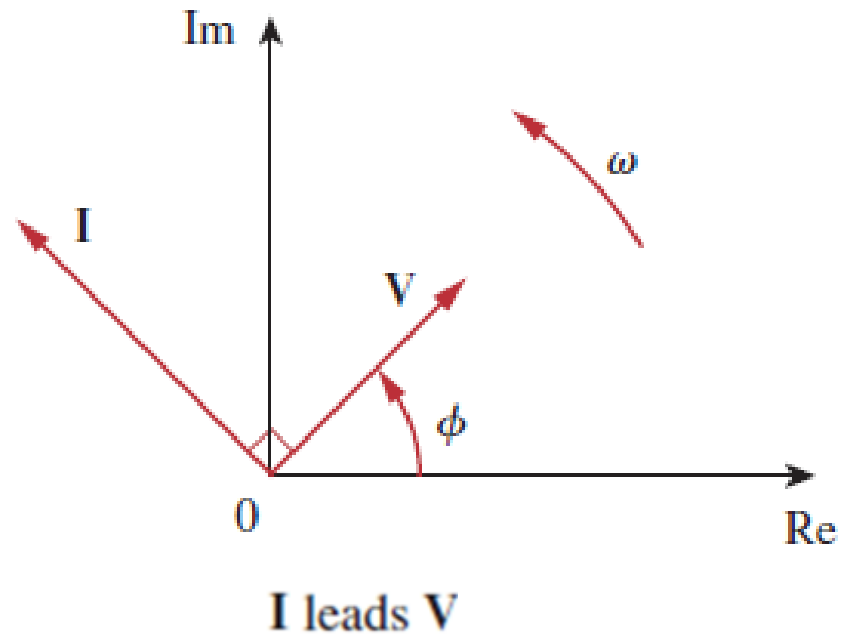
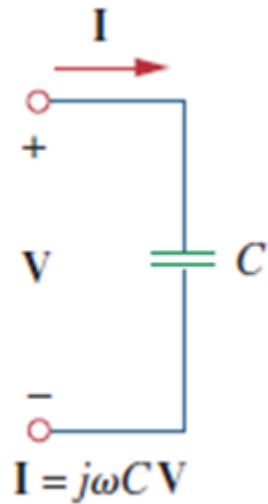
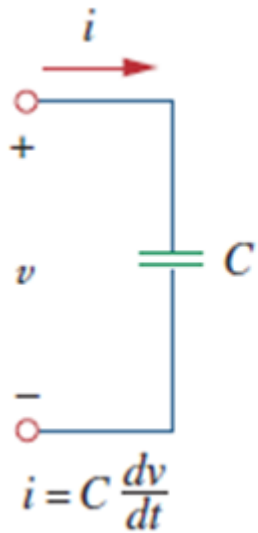
Phasor relationship for circuit elements

INDUCTOR



Phasor relationship for circuit elements

CAPACITOR



Phasor relationship for circuit elements

Voltage – Current relationship in circuit elements

Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$
L	$v = L \frac{di}{dt}$	$V = j\omega LI$
C	$i = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$

Impedance

The impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I , measured in ohms.

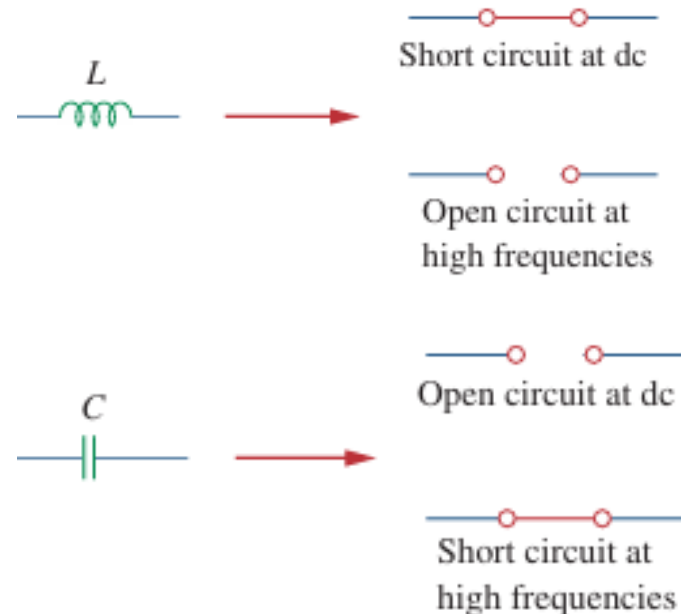
$$V = RI, \quad V = j\omega LI, \quad V = \frac{I}{j\omega C}$$

$$\frac{V}{I} = R, \quad \frac{V}{I} = j\omega L, \quad \frac{V}{I} = \frac{1}{j\omega C}$$

$$Z = \frac{V}{I} \quad \text{or} \quad V = ZI$$

Z is a frequency-dependent quantity known as impedance, measured in ohms.

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$



Phasor relationship for circuit elements

As a complex quantity, the impedance may be expressed in rectangular form as

$$\mathbf{Z} = R + jX$$

The impedance may also be expressed in polar form as

$$\mathbf{Z} = |\mathbf{Z}| \angle \theta$$

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta$$

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

$$R = |\mathbf{Z}| \cos \theta, \quad X = |\mathbf{Z}| \sin \theta$$

Admittance

The admittance Y is the reciprocal of impedance, measured in Siemens (S).

The admittance Y of an element (or a circuit) is the ratio of the phasor current through it to the phasor voltage across it.

$$Y = \frac{1}{Z} = \frac{I}{V}$$

$$Y = G + jB$$

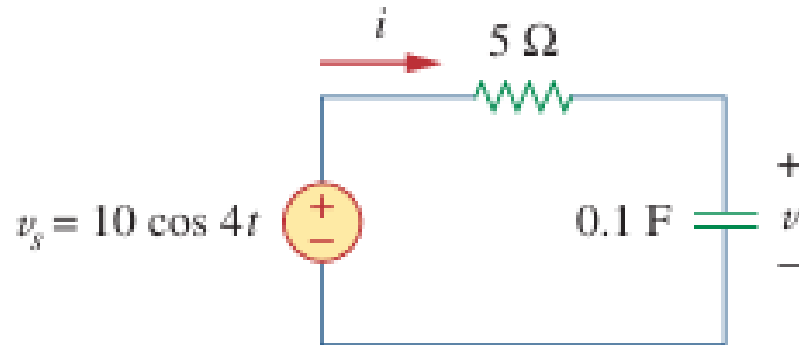
$$G + jB = \frac{1}{R + jX}$$

$$G + jB = \frac{1}{R + jX} \cdot \frac{R - jX}{R - jX} = \frac{R - jX}{R^2 + X^2}$$

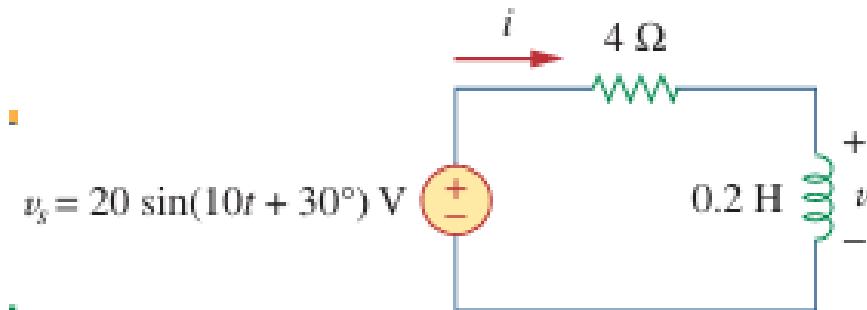
$$G = \frac{R}{R^2 + X^2}, \quad B = -\frac{X}{R^2 + X^2}$$

Exercise

1. Find $v(t)$ and $i(t)$ in the circuit.



2. Find $v(t)$ and $i(t)$ in the circuit.



Next class: Kirchhoff's laws in frequency domain;
instantaneous and average power; complex power